

PayTracka

API Contract

Version 1.0 — All endpoints, request/response shapes, and integration rules

Auth Service Base URL	https://paytracka-auth.railway.app
Core API Base URL	https://paytracka-api.railway.app
Interswitch Sandbox	https://sandbox.interswitchng.com
Interswitch Live	https://interswitchng.com
Anthropic API	https://api.anthropic.com/v1/messages

1. Authentication Rules

All protected endpoints require a JWT Bearer token in the Authorization header:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Security: Tokens are issued on login and expire after 24 hours. The frontend stores tokens in Supabase session — not localStorage.

2. Standard Response Format

Every API response — both services — follows this envelope:

```
// Success
{ "success": true, "data": { ... } }

// Error
{ "success": false, "error": "ERROR_CODE", "message": "Human readable message" }
```

3. Auth Service Endpoints (Node.js)

Method	Endpoint	Auth	Description
POST	/api/auth/register	No	Create user account with optional BVN verification
POST	/api/auth/login	No	Login — returns JWT token and user object
GET	/api/auth/me	JWT	Get current authenticated user profile
POST	/api/auth/logout	JWT	Invalidate current session token
POST	/api/payments/generate-link	JWT	Generate Interswitch payment link for an invoice
GET	/api/payments/callback	No	Interswitch post-payment redirect handler
POST	/api/payments/webhook	No	Interswitch transaction confirmation webhook

3.1 POST /api/auth/register

Request body:

Field	Type	Required	Description
full_name	string	Yes	User's full name
email	string	Yes	Unique email address
password	string	Yes	Minimum 8 characters
phone	string	No	Nigerian phone number e.g. 08012345678
bvn	string	No	11-digit BVN — triggers Interswitch Identity verification

Success response (201):

```
{
  "success": true,
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "user": {
      "id": "uuid",
      "full_name": "Oluchi Ruth",
      "email": "oluchi@email.com",
      "bvn_verified": true
    }
  }
}
```

3.2 POST /api/payments/generate-link

Request body:

Field	Type	Required	Description
invoice_id	string (uuid)	Yes	The invoice to generate a link for

Field	Type	Required	Description
amount	number	Yes	Amount in Naira — system converts to kobo automatically
client_email	string	Yes	Client email for Interswitch payment page
client_name	string	Yes	Client name shown on Interswitch payment page
description	string	No	Invoice description shown on payment page

Success response (200):

```
{
  "success": true,
  "data": {
    "payment_link": "https://newwebpay.qa.interswitchng.com/collections/w/pay?...",
    "txn_ref": "PT-a1b2c3d4-1718900000",
    "expires_at": "2025-06-22T09:41:00Z"
  }
}
```

4. Core API Endpoints (FastAPI)

Method	Endpoint	Auth	Description
GET	/api/invoices	JWT	List all invoices — supports ?status= filter
POST	/api/invoices	JWT	Create a new invoice
GET	/api/invoices/{id}	JWT	Get single invoice detail
PUT	/api/invoices/{id}	JWT	Update invoice fields
DELETE	/api/invoices/{id}	JWT	Delete an invoice
GET	/api/guard/alerts	JWT	Get all Smart Guard alerts (overdue, due soon, paid)
POST	/api/guard/remind/{id}	JWT	Generate AI reminder message for a client
GET	/api/insights/summary	JWT	Monthly income summary and key metrics
GET	/api/insights/clients	JWT	Client reliability scores
GET	/api/insights/monthly	JWT	Month-by-month income breakdown
GET	/health	No	Service health check

4.1 POST /api/invoices — Create Invoice

Request body:

Field	Type	Required	Description
client_name	string	Yes	Name of the client
client_email	string	Yes	Client email for payment link generation
amount	number	Yes	Invoice amount in Naira
due_date	string (ISO date)	Yes	Due date e.g. 2025-06-20
payment_method	string	Yes	bank_transfer or pay_link
status	string	No	pending paid overdue — default: pending
description	string	No	Work description — can be AI-generated

Success response (201):

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "invoice_ref": "INV-006",
    "client_name": "Joanna Tebadda",
    "client_email": "joanna@email.com",
    "amount": 150000,
    "due_date": "2025-01-14",
    "payment_method": "pay_link",
```

```
    "status": "overdue",
    "description": "UI design for landing page",
    "created_at": "2025-01-01T09:00:00Z"
  }
}
```

4.2 GET /api/guard/alerts — Smart Guard

Response:

```
{
  "success": true,
  "data": {
    "overdue": [
      { "invoice_ref": "INV-001", "client_name": "Joanna Tebadda",
        "amount": 150000, "days_overdue": 12 }
    ],
    "due_soon": [
      { "invoice_ref": "INV-002", "client_name": "Glory Clement",
        "amount": 80000, "days_until_due": 2 }
    ],
    "recently_paid": [
      { "invoice_ref": "INV-004", "client_name": "Amara Osei Studio",
        "amount": 65000, "paid_at": "2025-01-12T10:00:00Z" }
    ]
  }
}
```

4.3 GET /api/insights/summary

Response:

```
{
  "success": true,
  "data": {
    "total_earned": 65000,
    "pending_total": 200000,
    "overdue_total": 270000,
    "invoice_count": 5,
    "avg_payment_days": 8.4,
    "expected_14_days": 120000
  }
}
```

5. Interswitch API Reference

5.1 OAuth Token

Property	Value
Endpoint	POST https://sandbox.interswitchng.com/passport/oauth/token
Auth header	Basic Base64(CLIENT_ID:SECRET_KEY)
Body	grant_type=client_credentials
Token expiry	86400 seconds (24 hours)
Live URL	https://interswitchng.com/passport/oauth/token

5.2 Web Checkout Payment Link

Parameter	Required	Description
merchant_code	Yes	From business.quickteller.com Developer Tools
pay_item_id	Yes	From business.quickteller.com Developer Tools
txn_ref	Yes	Your unique transaction reference — save in DB
amount	Yes	Amount in KOBO — multiply Naira by 100
currency	Yes	566 for NGN (ISO 4217 numeric code)
cust_email	Yes	Client's email address
site_redirect_url	Yes	Your backend callback URL after payment

5.3 Transaction Verification

Property	Value
Endpoint	GET https://qa.interswitchng.com/collections/api/v1/gettransaction.json
Parameters	merchantcode, transactionreference, amount (in kobo)
Success code	ResponseCode: '00' = Approved
Live URL	https://webpay.interswitchng.com/collections/api/v1/gettransaction.json

5.4 BVN Identity Verification

Property	Value
Endpoint	POST https://sandbox.interswitchng.com/api/v1/identity/bvn
Auth	Bearer token from OAuth
Body	{ bvn: '22222222226' }
Response	{ verified: true/false, firstName, lastName }

Property	Value
Test BVN	22222222226
Credentials from	developer.interswitchgroup.com — API Marketplace

5.5 Interswitch Test Cards

Card Number	Expiry	CVV	Result
5061 0300 0000 0000 04	12/26	123	Approved — use for demo
5061 0300 0000 0000 13	12/26	123	Declined
4000 0000 0000 0002	12/26	123	3D Secure required

6. Error Codes Reference

Error Code	HTTP Status	Description
AUTH_INVALID_CREDENTIALS	401	Wrong email or password
AUTH_TOKEN_EXPIRED	401	JWT expired — user must log in again
INVOICE_NOT_FOUND	404	Invoice does not exist or belongs to another user
PAYMENT_LINK_FAILED	502	Interswitch API call failed
BVN_VERIFY_FAILED	422	BVN could not be verified — registration still proceeds
VALIDATION_ERROR	422	Request body missing required fields
AI_GENERATION_FAILED	503	Anthropic API call failed — show fallback message

This API contract is the source of truth for all PayTracka integrations. Lock this document before coding begins. Any endpoint changes must be agreed by all team members.